Task 1: Research on the Data Analyst Role

📄 **Page 1: Introduction + Responsibilities**

🔹 **What is a Data Analyst?**

A **Data Analyst** is a professional who collects, processes, and interprets data to help organizations make informed decisions. They transform raw data into meaningful insights using analytical tools and techniques.

🔹 **Key Responsibilities**

1. **Data Collection**
   - Gather data from databases, APIs, surveys, and other sources
2. **Data Cleaning**
   - Remove errors, duplicates, and inconsistencies
3. **Data Analysis**
   - Use statistical methods to identify patterns and trends
4. **Data Visualization**
   - Create charts, dashboards, and reports
5. **Reporting**
   - Present insights to stakeholders for decision-making
6. **Problem Solving**
   - Answer business questions using data

🔹 **Importance in Tech Industry**

- Helps companies make **data-driven decisions**
- Improves **business efficiency**
- Identifies **market trends**
- Supports **AI & Machine Learning models**
- Reduces risks and increases profitability

🔹 **Popular Tools Used by Data Analysts**

Tool	Purpose
Excel	Basic data analysis
SQL	Database querying
Python	Advanced analysis & automation
R	Statistical analysis
Power BI / Tableau	Data visualization
Google Sheets	Cloud-based data handling

🔹 Key Skills Required

1. Technical Skills

- SQL
- Python (Pandas, NumPy)
- Excel
- Data Visualization

2. Analytical Skills

- Logical thinking
- Problem-solving
- Statistical understanding

3. Soft Skills

- Communication
- Storytelling with data
- Attention to detail

🔹 Example Workflow

1. Collect Data
2. Clean Data
3. Analyze Data

4. Visualize Results
5. Share Insights

📄 **Page 3: Career + Conclusion**

🔹 **Career Opportunities**

- Data Analyst
- Business Analyst
- Data Scientist (after upskilling)
- Data Engineer

🔹 **Salary (India – Approx)**

- Fresher: ₹3–6 LPA
- Mid-level: ₹6–12 LPA
- Experienced: ₹12+ LPA

🔹 **Why Choose Data Analyst?**

- High demand in every industry
- Good salary growth
- Entry-friendly for freshers
- Gateway to advanced tech roles

🔹 **Conclusion**

The role of a Data Analyst is crucial in today's digital world. By turning data into actionable insights, analysts help businesses grow, innovate, and stay competitive.

❓ **Q&A SECTION (MANDATORY)**

Q1. What does a Data Analyst do?

A Data Analyst collects, processes, and analyzes data to help organizations make better decisions.

Q2. Which tools are most important?

SQL, Excel, Python, and visualization tools like Tableau or Power BI.

Q3. Is coding required?

Basic coding (Python/SQL) is very helpful but not always mandatory at beginner level.

Q4. Difference between Data Analyst and Data Scientist?

- Analyst: Works on insights & reporting
- Scientist: Builds predictive models

Q5. Can a fresher become a Data Analyst?

Yes, with skills in Excel, SQL, and basic Python.